

 Assessment Playback**Assistant**

00:00

Which of the following statements accurately describe properties of Breadth-First Search (BFS), Depth-First Search (DFS), Best-First Search (BeFS), and A* search?

A. BFS guarantees finding the shortest path in unweighted graphs (optimality) but can have high memory usage. DFS is generally faster than BFS when the goal is deep but does not guarantee optimality. A* search guarantees optimality if the heuristic is admissible and consistent, and it typically balances memory and speed.

B. BFS always finds the goal fastest if it exists. DFS is guaranteed to be optimal. BeFS prioritizes exploring nodes closer to the goal based on a cost function. A* search is a form of DFS that uses a heuristic.

C. DFS guarantees finding the shortest path. BFS can have very low memory usage. BeFS guarantees optimality. A* search is primarily focused on speed regardless of path optimality.

D. BFS is optimal and memory-efficient. DFS is guaranteed to find the goal quickly. BeFS is a blind search algorithm. A* search is a randomized search algorithm.

User 00:00

A

Assistant

00:00

Consider a scenario where you need to find the shortest path in a large, unweighted graph with a known goal state. You are evaluating different search algorithms. Which of the following statements accurately describes the performance characteristics and

suitability of BFS, DFS, BeFS, and A* in this context?

A. BFS guarantees finding the shortest path (optimality) in an unweighted graph and has predictable, generally high memory usage due to storing all nodes at the current depth. DFS may not find the shortest path and has lower memory usage but can get stuck in deep branches.

B. A* search, with an admissible and consistent heuristic, will find the shortest path and is generally faster than BFS on average due to its informed exploration, but its memory usage can be comparable to BFS in worst-case scenarios.

C. Best-First Search (BeFS) prioritizes nodes based on a heuristic function alone, making it potentially faster than BFS if the heuristic is good, but it does not guarantee optimality. DFS is always the most memory-efficient choice regardless of the graph structure.

D. In an unweighted graph, BFS and A* (with a zero heuristic) are equivalent in their guarantee of finding the shortest path. DFS is always preferred for its speed in such scenarios as it explores branches until an optimal solution is found.

User 00:00

A, B, C

Assistant

00:00

You are using the A* search algorithm to find a path from a start node 'Start' to a goal node 'Goal'. The cost to reach a node from the start is denoted by $g(n)$, and the heuristic estimate from node n to the goal is $h(n)$. The $f(n)$ value is calculated as $f(n) = g(n) + h(n)$. Consider the following state of the open list (priority queue ordered by $f(n)$ value) during the A* search:

Node	$g(n)$	$h(n)$	$f(n)$
A	4	3	7

B | 6 | 1 | /

C | 2 | 6 | 8

D | 5 | 2 | 7

- A. Node A could be expanded next.
- B. Node B could be expanded next.
- C. Node C could be expanded next.
- D. Node D could be expanded next.

User 00:00

A

Assistant

00:00

In a search algorithm like A*, what does it mean for a heuristic function 'h(n)' to be considered 'admissible'?

- A. The heuristic always overestimates the true cost to reach the goal.
- B. The heuristic always underestimates or is exactly equal to the true cost to reach the goal.
- C. The heuristic provides the exact cost to reach the goal.
- D. The heuristic is based on the path cost already incurred to reach the current node.

User 00:00

B

Assistant

00:00

In the context of planning problems like optimizing the loading of a truck with various items (each with different weights and volumes), which of the following best describes the distinction between path cost and solution quality when using search algorithms?

- A. Path cost refers to the computational effort to find a solution, while solution quality refers to how efficiently the truck is loaded (e.g., maximizing volume utilization).
- B. Path cost is always the primary metric to minimize, and solution quality is a secondary consideration.
- C. Solution quality is directly proportional to the number of steps taken in the search, and path cost is irrelevant.

irrelevant.

D. Path cost and solution quality are interchangeable terms when discussing planning problems.

User 00:00

A

Assistant

00:00

Consider a robotic arm tasked with assembling a complex product. The state of the arm can be represented by its joint angles, the position of the product components, and whether each component is correctly placed. Actions include moving specific joints, grasping a component, and placing a component. The 'path cost' in this scenario might be the total energy consumed by the motors to move the arm. However, simply minimizing energy might lead to a very slow assembly or even a suboptimal final product placement (e.g., a component might be in the correct location but angled slightly wrong). Which of the following best describes the distinction being illustrated and a potential approach to address it?

A. The distinction is between path cost (energy) and solution quality (correct assembly and efficiency). An admissible heuristic for A* search could estimate the remaining assembly steps and component placement accuracy, prioritizing states that are both energy-efficient and closer to a complete, correct assembly.

B. Path cost is irrelevant; only solution quality matters. A Breadth-First Search (BFS) is ideal because it guarantees finding the solution with the fewest actions, which directly translates to the highest solution quality.

C. The problem illustrates that path cost and solution quality are always inversely proportional. Therefore, we should always prioritize minimizing path cost to ensure the highest solution quality.

D. The primary challenge is designing a state representation that includes all

possible assembly errors. Once this is done, any search algorithm will inherently optimize for both path cost and solution quality simultaneously.

User 00:00
A

Assistant 00:00
Assessment complete. Thank you.

Recording

Evaluation

Compare Search Algorithms

Grade B

MULTIPLE CHOICE

Which of the following statements accurately describe properties of Breadth-First Search (BFS), Depth-First Search (DFS), Best-First Search (BeFS), and A* search?

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- B. BFS always finds the goal fastest if it exists. DFS is guaranteed to be optimal. BeFS prioritizes exploring nodes closer to the goal based on a cost function. A* search is a form of DFS that uses a heuristic.
- C. DFS guarantees finding the shortest path. BFS can have very low memory usage. BeFS guarantees optimality. A* search is primarily focused on speed regardless of path optimality.
- D. BFS is optimal and memory-efficient. DFS is guaranteed to find the goal quickly. BeFS is a blind search algorithm. A* search is a randomized search algorithm.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A

EXPLANATION

Option A correctly identifies the key properties of each search algorithm, including their strengths and weaknesses regarding goal finding, optimality, and memory/speed characteristics.

SCORE

5

MULTIPLE CHOICE

Consider a scenario where you need to find the shortest path in a large, unweighted graph with a known goal state. You are evaluating different search algorithms. Which of the following statements accurately describes the performance characteristics and suitability of BFS, DFS, BeFS, and A* in this context?

- A. BFS guarantees finding the shortest path (optimality) in an unweighted graph and has predictable, generally high memory usage due to storing all nodes at the current depth. DFS may not find the shortest path and has lower memory usage but can get stuck in deep branches.
- B. A* search, with an admissible and consistent heuristic, will find the shortest path and is generally faster than BFS on average due to its informed exploration, but its memory usage can be comparable to BFS in worst-case scenarios.
- C. Best-First Search (BeFS) prioritizes nodes based on a heuristic function alone, making it potentially faster than BFS if the heuristic is good, but it does not guarantee optimality. DFS is always the most memory-efficient choice regardless of the graph structure.
- D. In an unweighted graph, BFS and A* (with a zero heuristic) are equivalent in their guarantee of finding the shortest path. DFS is always preferred for its speed in such scenarios as it explores branches until an optimal solution is found.

STUDENT ANSWER

A, B, C

CORRECT ANSWER(S)

A, B

EXPLANATION

BFS guarantees optimality in unweighted graphs and has high memory use. A* with zero heuristic is same as BeFS and also BFS in an unweighted graph. DFS is not guaranteed to be optimal and its speed is highly dependent on graph structure.

SCORE

2.5

Evaluate and Apply Heuristics

Grade C

MULTIPLE CHOICE

You are using the A* search algorithm to find a path from a start node 'Start' to a goal node 'Goal'. The cost to reach a node from the start is denoted by $g(n)$, and the heuristic estimate from node n to the goal is $h(n)$. The $f(n)$ value is calculated as $f(n) = g(n) + h(n)$. Consider the following state of the open list (priority queue ordered by $f(n)$ value) during the A* search:

Node | $g(n)$ | $h(n)$ | $f(n)$

----|-----|-----|-----

A | 4 | 3 | 7

B | 6 | 1 | 7

C | 2 | 6 | 8

D | 5 | 2 | 7

A. Node A could be expanded next.

B. Node B could be expanded next.

C. Node C could be expanded next.

D. Node D could be expanded next.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A, B, D

EXPLANATION

When multiple nodes have the same lowest $f(n)$ value, A^* might expand any of them. In this case, A, B, and D all have the lowest $f(n)$ of 7.

SCORE

1.6667

MULTIPLE CHOICE

In a search algorithm like A^* , what does it mean for a heuristic function ' $h(n)$ ' to be considered 'admissible'?

A. The heuristic always overestimates the true cost to reach the goal.

B. The heuristic always underestimates or is exactly equal to the true cost to reach the goal.

C. The heuristic provides the exact cost to reach the goal.

D. The heuristic is based on the path cost already incurred to reach the current node.

STUDENT ANSWER

B

CORRECT ANSWER(S)

B

EXPLANATION

An admissible heuristic never overestimates the cost to reach the goal, ensuring that A^* will find the optimal solution.

SCORE

5

MULTIPLE CHOICE

In the context of planning problems like optimizing the loading of a truck with various items (each with different weights and volumes), which of the following best describes the distinction between path cost and solution quality when using search algorithms?

- A. Path cost refers to the computational effort to find a solution, while solution quality refers to how efficiently the truck is loaded (e.g., maximizing volume utilization).
- B. Path cost is always the primary metric to minimize, and solution quality is a secondary consideration.
- C. Solution quality is directly proportional to the number of steps taken in the search, and path cost is irrelevant.
- D. Path cost and solution quality are interchangeable terms when discussing planning problems.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A

EXPLANATION

Path cost in search algorithms typically refers to the cost associated with traversing the search space (e.g., number of nodes expanded or computation time), whereas solution quality in a planning problem like truck loading refers to how well the final configuration meets the objectives (e.g., maximizing the value or volume of items loaded).

SCORE

5

MULTIPLE CHOICE

Consider a robotic arm tasked with assembling a complex product. The state of the arm can be represented by its joint angles, the position of the product components, and whether each component is correctly placed. Actions include moving specific joints, grasping a component, and placing a component. The 'path cost' in this scenario might be the total energy consumed by the motors to move the arm. However, simply minimizing energy might lead to a very slow assembly or even a suboptimal final product placement (e.g., a component might be in the correct location but angled slightly wrong). Which of the following best describes the distinction being illustrated and a potential approach to address it?

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C. The problem illustrates that path cost and solution quality are always inversely proportional. Therefore, we should always prioritize minimizing path cost to ensure the highest solution quality.

D. The primary challenge is designing a state representation that includes all possible assembly errors. Once this is done, any search algorithm will inherently optimize for both path cost and solution quality simultaneously.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A

EXPLANATION

Option A correctly identifies the trade-off and proposes a relevant heuristic for A* search to balance energy efficiency (path cost) with assembly correctness and speed (solution quality).

SCORE

5

79.6%

Good

Overall Score

100%

High

Honesty Score

Assessment completed across 3 skills. Skill scores: Compare Search Algorithms: 73.8/100, Evaluate and Apply Heuristics: 64.9/100, Model Planning Problems as State-Space Search: 100.0/100.